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The Impact of Mixed Wettability on Pore-Scale Fluid Displacement Dynamics in Microfluidic Models

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Abstract

This study explores the role of mixed wettability in influencing fluid displacement behaviors at the pore scale, which is a critical yet underexplored aspect of subsurface fluid flow. By replicating natural porous media conditions in microfluidic devices, we investigated how wettability heterogeneity influences injection time, displacement patterns, dynamic pressure, and sweep efficiency. These findings provide insights relevant to applications such as enhanced oil recovery, and CO₂ sequestration, among others. A series of microfluidic devices featuring rock-like structures with controlled wettability variations were fabricated using photolithography and molecular vapor deposition (MVD) of perfluorodecyltrichlorosilane (FDTs). Predominantly hydrophobic surfaces were selectively modified in narrow pore throats based on pore size distribution, effectively mimicking the mixed wettability characteristics commonly observed in natural rock formations. Experimental setups included water injection into air-saturated micromodels under capillary-dominated conditions, with real-time high-resolution imaging capturing pore-scale displacement patterns. A range of wettability conditions including fully hydrophobic, fully hydrophilic, and mixed wet, was systematically tested to assess the influence of wettability on fluid dynamics. The findings reveal that wettability significantly impacts fluid displacement dynamics, with distinct behaviors observed for different wettability conditions. Dynamic pressures were inversely proportional to the fraction of hydrophilic zones, reflecting the influence of wetting pore throats on invasion pressure and pressure curve shifts. Saturation trends and pore-filling sequences varied across wettability conditions, highlighting the critical role of localized wettability changes in redirecting fluid flow. Mixed wettability introduced a balance between capillary and viscous forces, creating unique displacement behaviors not observed in uniform-wet systems. These findings challenge conventional assumptions by demonstrating that mixed-wet systems exhibit unique and complex dynamics rather than serving as intermediates between uniform extremes. This study employs high-resolution microfluidic models with pore-size-correlated mixed wettability distributions, establishing a robust framework for analyzing fluid dynamics in heterogeneous porous media. The results advance our understanding of multiphase flow in mixed-wet systems, offering practical insights for optimizing fluid management in subsurface energy applications.

Introduction

Multiphase flow within porous media is a fundamental phenomenon integral to a variety of natural and engineering applications, such as hydrocarbon extraction (Lake et al., 2014), geological carbon sequestration (Al-Khdheawi et al., 2017), hydrogen storage (H. Zhang et al., 2024), and soil water infiltration (Cueto-Felgueroso and Juanes, 2008). Immiscible displacement, which describes the replacement of one fluid within a confined pore space by another, is largely governed by wettability (Anderson, 1986). Despite the significant impact of wettability on fluid-solid interactions, its intricate physics remain a challenge in accurately predicting fluid behavior at both microscopic and macroscopic scales.

Mixed wettability widely exists in natural oil-bearing reservoirs (Tiab and Donaldson, 2015), characterized by distinct wetting states in different areas of the porous medium. This definition marks a fundamental difference from neutral or intermediate wettability when the medium shows no preference for a specific fluid. Mixed wettability represents a form of wettability alteration that is location-controlled and correlated to pore size distribution (Liang et al., 2023; Wang et al., 2018), as illustrated schematically in Fig. 1. During the primary drainage process of a water-saturated reservoir, oil typically infiltrates the large pores, while some water persists in smaller pores due to high capillarity. The process is accompanied by the deposition of surface-active hydrocarbon components (e.g., asphaltene and resins) onto the walls of large pores due to prolonged exposure to the oil phase (Kovscek et al., 1993; Patzek et al., 2022). While mixed wettability is a naturally prevalent phenomenon, its influence on pore-scale immiscible fluid displacement remains inadequately understood.

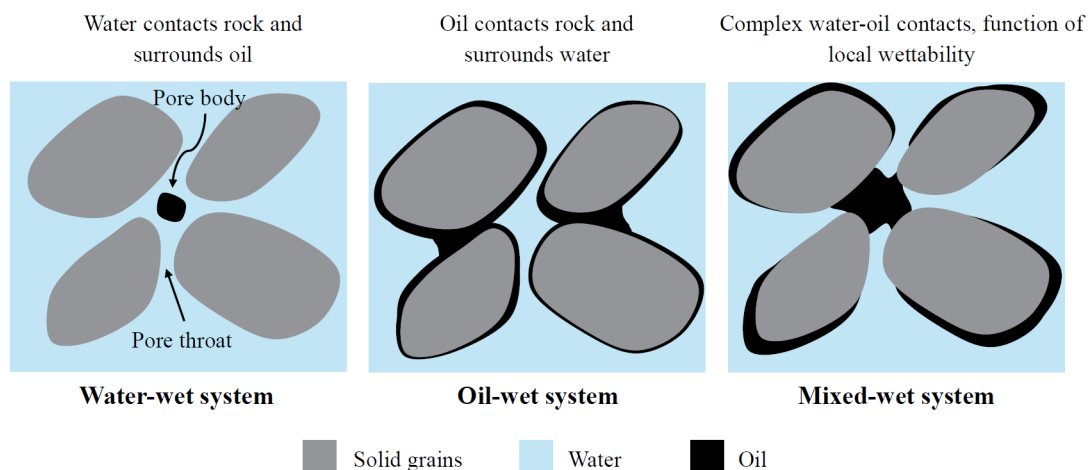


Figure 1—An idealized schematic illustrating fluid distribution under different wettability conditions: Water-wet conditions, characterized by hydrophilic surfaces where water contacts the rock surface, resulting in oil being trapped at the pore centers. Oil-wet conditions, characterized by hydrophobic surfaces where oil contacts the rock surface, causing water to be trapped at the pore centers. Mixed-wet conditions, displaying a complex distribution where localized wettability variations result in oil occupying large pores and contact surfaces, while water resides in the narrower pores and pore throats.

Several studies have examined the impact of mixed wettability on fluid flow using altered core samples (Alhosani et al., 2023; Rücker et al., 2019). However, challenges in accurately characterizing the in-situ wetting properties of core samples, and in evaluating the localized effects of mixed wettability, have constrained efforts to draw definitive conclusions. Microfluidics, as analogue systems, offer a complementary approach to overcoming these limitations. By enabling rapid, precise, and real-time analyses, while replicating representative pore-size models, microfluidic systems provide valuable insights into the influence of mixed wettability on fluid-fluid displacement. Despite their potential, only a few recent microfluidic studies have investigated the role of mixed wettability, employing either numerical simulations (Ahmadi-Falavarjani et al., 2024; Irannezhad et al., 2023a; S. Zhang et al., 2024) or experimental approaches

with simplified models (Chang et al., 2020; Irannezhad et al., 2023b; Sun and Torres-Verdín, 2023). Nevertheless, replicating representative mixed wettability that accurately reflects its natural orientation remains a significant challenge. A significant challenge lies in replicating realistic mixed wettability that accurately mirrors natural systems. Addressing this, AlOmier recently introduced an innovative method for creating spatially distributed mixed-wet surfaces in microfluidics using photolithography and thin-film deposition (AlOmier et al., 2024a). This approach holds substantial potential for generating well-controlled mixed-wet surfaces, paving the way for advancements in various applications.

This study examines the role of mixed wettability in immiscible fluid-fluid displacement and invasion patterns using advanced microfluidic devices. A novel technique combining photolithography and molecular vapor deposition (MVD) of perfluorodecyltrichlorosilane (FDTS) was employed to create pore-scale selective wettability coatings. To facilitate a systematic analysis, identical microfluidic chips with hydrophilic, hydrophobic, and three distinct mixed-wet configurations were developed. Displacement experiments provided detailed insights into the effects of wettability variations.

Methodology

Silicon-based microfluidic devices were fabricated using advanced semiconductor photolithography and dry-etching techniques to simulate pore networks of oil-bearing rocks exhibiting both single and mixed wettability. The microfluidic porous medium replicates the pore network of a reservoir rock, derived from a thin-section image of a sandstone core. **Fig. 2a**, presents a digitalized binary image with an edge length of 3.3 mm, highlighting the rock features. In this image, white areas represent pore spaces, while black regions correspond to solid grains. The processed image was transformed into a micromodel layout, as shown in **Fig. 2b**, featuring inlet and outlet channels. The pore network is displayed in purple, and the solid components are shown in white. The resulting micromodel is characterized by an average pore radius of 17 μm , and a minimum feature size of 3 μm . Displacement experiments were conducted by injecting water into air-saturated microfluidic chips at a constant flow rate, designed to emphasize capillary-dominated conditions and magnify the impact of wettability as depicted in **Fig. 2c**. High-resolution imaging of water distributions offered direct observations of pore-scale displacement patterns.

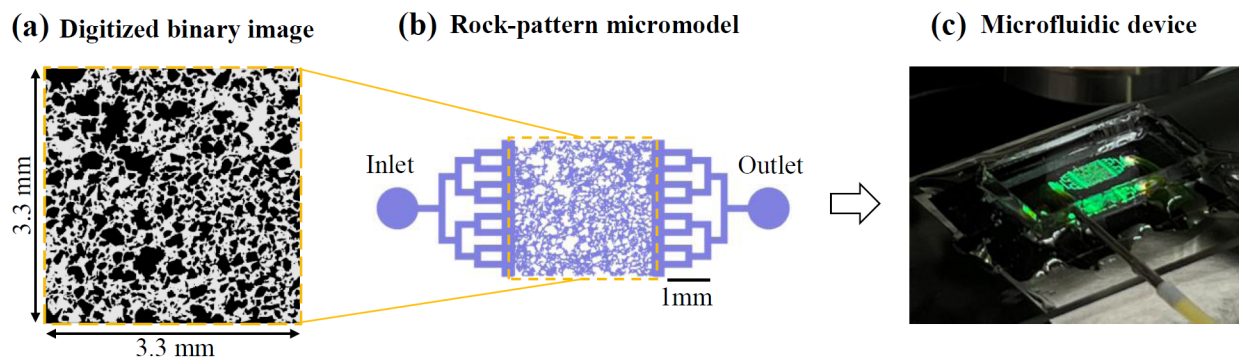


Figure 2—Design and fabrication of micromodels with rock-patterned porous media: (a) Binarized black-and-white image of a sandstone thin section showing a flow matrix with an edge length of 3.3 mm. (b) Micromodel layout design, depicting the pore network in purple and the solid grains in white. (c) Fabricated microfluidic device in use during a displacement experiment.

Five micromodels with identical pore-network structures but varying wettability properties were fabricated, including hydrophilic, hydrophobic, and three mixed-wettability configurations. A novel approach combining a second layer of photolithography with thin film deposition was utilized to create selective pore-scale wettability coatings (AlOmier et al., 2024a). This method allowed for the precise alteration of wettability in narrow pore throats, effectively replicating the natural development of mixed wettability found in reservoir rocks. The substrate initially exhibited hydrophilic wettability,

which was subsequently modified to become hydrophobic by applying an FDTs coating through MVD. The hydrophilic and hydrophobic micromodels represented the two extreme cases of wettability, with bulk contact angles of $\theta=50^\circ$ and $\theta=113^\circ$, respectively, as illustrated in Fig. 3. To introduce mixed-wettability states, a selective hydrophobic FDTs coating was applied, modifying wettability at specific pore throats. Pore throats were ranked by size, with 200 identified for potential wettability alteration. Three mixed-wettability configurations were developed based on the percentage and spatial distribution of altered hydrophilic regions. In the Mix-High configuration, all 200 targeted pore throats were retained as hydrophilic, creating a high-density distribution of wettability alteration. The Mix-Med configuration included only the smallest 100 pore throats as hydrophilic, resulting in a moderate level of wettability alteration. Lastly, the Mix-Low configuration limited hydrophilicity to just the smallest 40 pore throats, representing the sparsest distribution of hydrophilic regions. This approach allowed for precise control over wettability distributions within the micromodel.

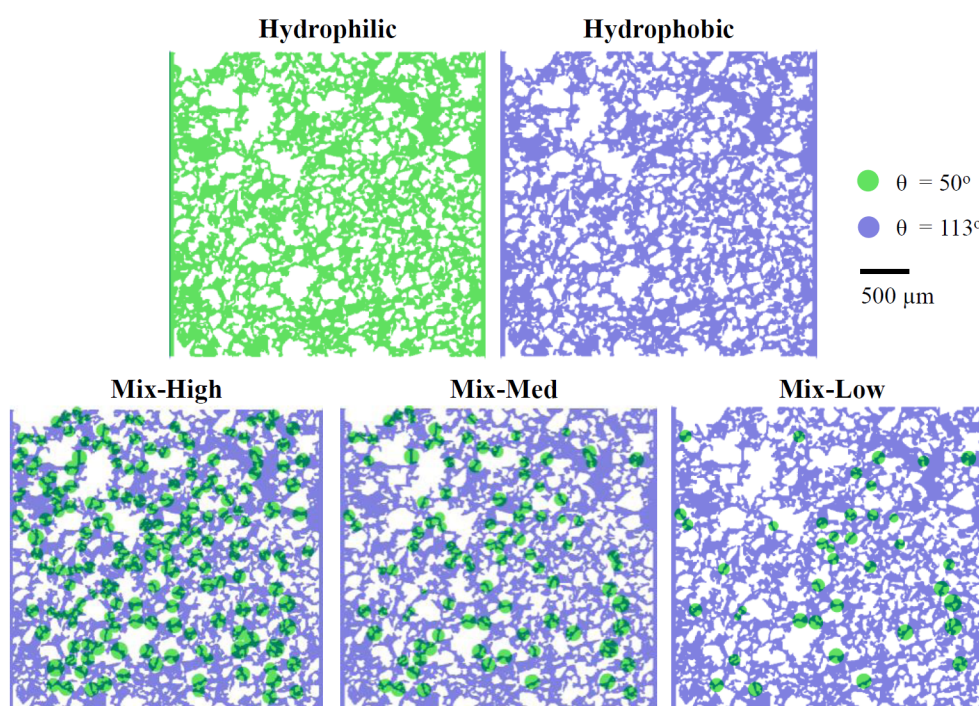


Figure 3—Micromodels demonstrating various wettability scenarios achieved through selective FDTs coating. The hydrophilic and hydrophobic micromodels exhibit bulk wettability values of $\theta=50^\circ$ (green) and $\theta=113^\circ$ (purple), respectively. Mixed-wettability configurations correspond to 100%, 50%, and 20% of the ranked pore-throat sizes. In the Mix-High case, all selected pore throats are included; in the Mix-Med and Mix-Low cases, only the smallest 100 and 40 pore throats are considered, respectively. The bulk surface of the mixed-wet micromodel is hydrophobic, while locally altered pore throats (green spots) are hydrophilic.

Results and Discussion

Impact on Spatial Displacement Patterns

The fluid-fluid displacement experiments under hydrophilic, hydrophobic, and mixed-wettability conditions (Mix-High, Mix-Med, Mix-Low) revealed distinct spatial displacement patterns (AlOmier et al., 2024b), as illustrated in Fig. 4. Given the fact that all five wettability scenarios are imprinted onto the same rock pattern, the pivotal impact of pore scale wettability alteration on fluid distribution becomes evident. In the hydrophilic scenario, water, as the wetting phase, experienced reduced capillary entry pressures due to favorable capillary forces. This reduction facilitated efficient pore invasion and enhanced overall displacement. Hydrophilic pore throats played a critical role by enabling cooperative filling of adjacent pore bodies, resulting in an interconnected water distribution throughout the matrix. In contrast, the

hydrophobic scenario exhibited significant resistance to water penetration. As a non-wetting phase, water had to overcome high capillary entry pressures to invade hydrophobic pore throats. This resistance led to preferential invasion of pores with the lowest entry pressures, while narrow pore throats acted as barriers, requiring water to build sufficient pressure for breakthrough. As a result, the displacement pattern exhibited reduced distribution across the porous matrix.

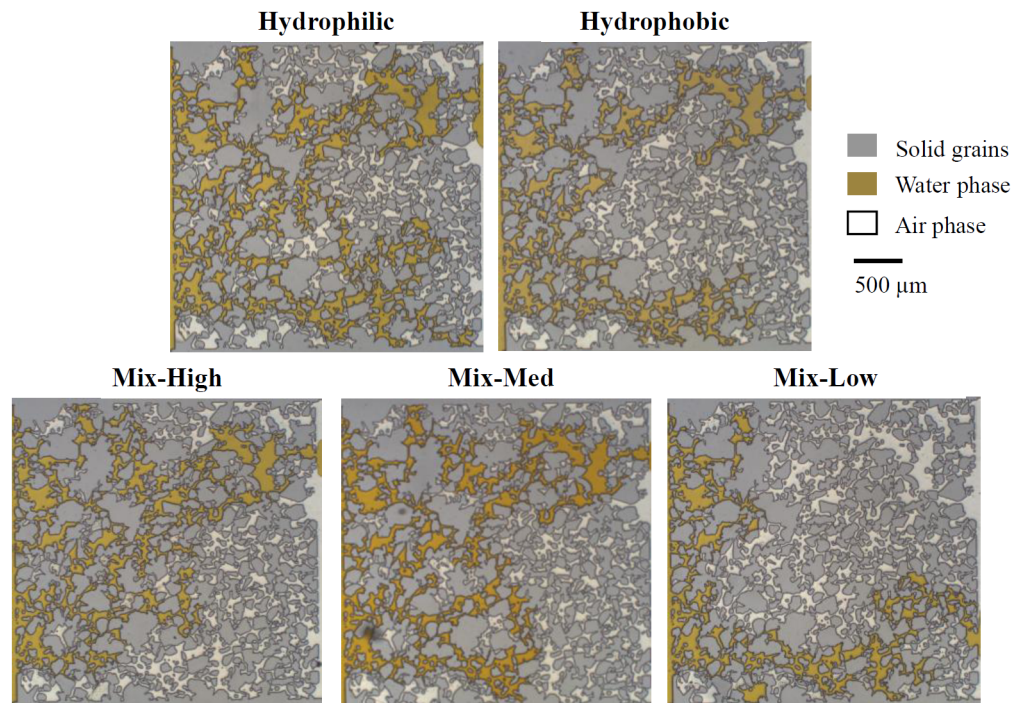


Figure 4—Experimental displacement patterns under various wetting conditions and configurations. These patterns correspond to the end of the experiment at the breakthrough when the invading fluid (yellow) displacing resident gas (white) reaches the right side of the flow matrix in microfluidic devices with rock-patterned features (grey). Local alterations pore-throats lead to unique displacement patterns between different cases.

Mixed-wettability systems exhibited displacement behaviors that were distinct from both hydrophilic and hydrophobic cases. The coexistence of hydrophilic and hydrophobic regions introduced complexity to the invasion patterns, significantly influencing displacement and sweep efficiency. Despite the predominantly hydrophobic nature of the porous structure, the presence of hydrophilic pore throats acted as preferential pathways, redirecting fluid flow and altering invasion patterns. This phenomenon was clearly observed in the Mix-High case, where a dense concentration of altered pore-throats was located in the middle of the matrix. Interestingly, this region was completely bypassed in the hydrophobic case. The impact of altered pore-throat distribution was also evident in the Mix-Med case, where fewer hydrophilic pore throats were more broadly dispersed. This expanded distribution facilitated enhanced fluid propagation throughout the porous matrix. The Mix-Low scenario, characterized by a predominantly hydrophobic matrix with only a few scattered hydrophilic pore throats, exhibited a markedly unique invasion pattern. Despite the minimal wettability alterations, these localized hydrophilic regions significantly influenced the flow, redirecting the fluid pathways and producing displacement behaviors distinct from all other scenarios.

These findings illustrate the profound impact of localized wettability adjustments on fluid flow behavior. The hydrophilic pore throats, even when present in small numbers, acted as critical nodes that dictated flow direction and connectivity. The displacement patterns observed across the mixed-wettability scenarios emphasize the interplay between capillary and viscous forces, regulated by the spatial arrangement and extent of wettability alterations. This interplay governs both the efficiency of fluid displacement and the overall sweep efficiency of the porous medium.

Impact on Dynamic Pressure and Fluid Saturation

Displacement experiments demonstrated unique pressure and saturation dynamics under hydrophilic, hydrophobic, and mixed-wettability conditions. Pressure drops and water saturations were monitored over time until breakthrough, defined as the moment when the displacement front reached the outlet of the porous matrix (**Fig. 5**). The results indicate an incomplete sweep of the matrix for all wettability scenarios, as evidenced by the saturation profiles at breakthrough. Pressure drops show a consistent increasing trend as water occupied progressively less accessible pore throats, reflecting rising hydraulic resistance. Sharp pressure reductions coincided with stabilization in saturation, marking the breakthrough event.

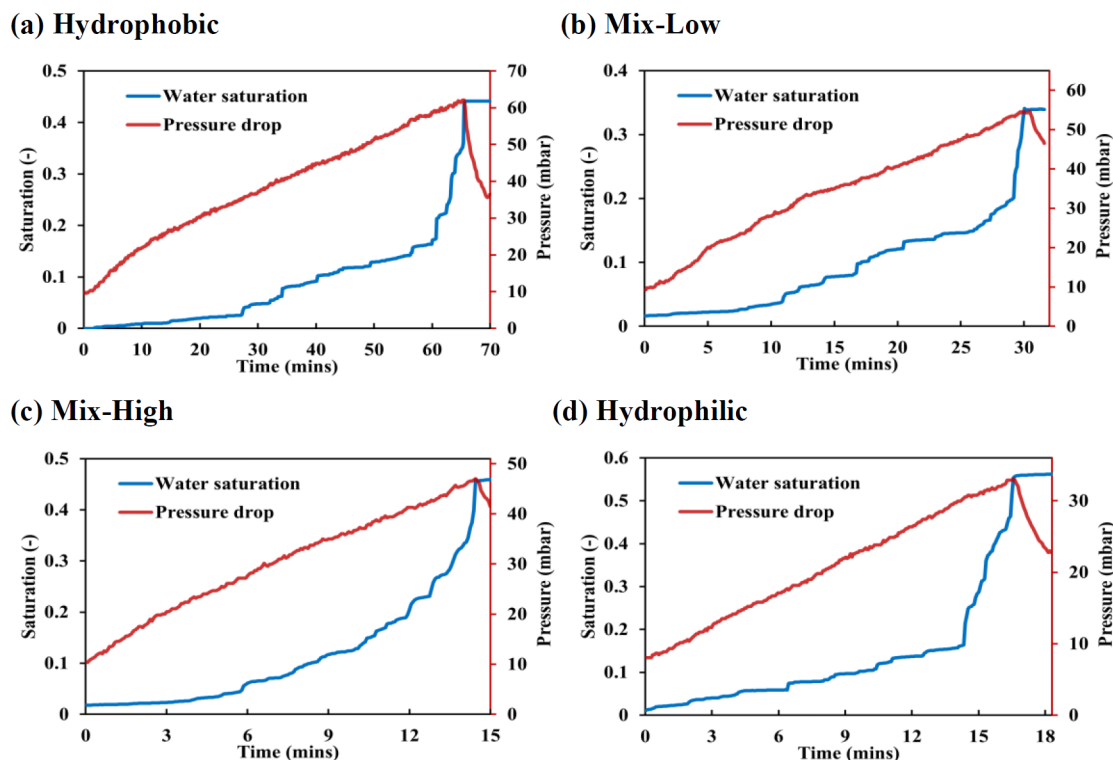


Figure 5—Measured pressure drop (red line) and water saturations (blue line) over time in the micromodel with wettability scenarios: Hydrophobic (a), Mix-Low (b), Mix-High (c), and Hydrophilic (d). The charts illustrate the increase in pressure drop corresponding to rising water saturation, followed by a sharp decline in pressure upon water breakthrough, indicating the establishment of fully connected pathways from the inlet to the outlet.

In the hydrophobic scenario (**Fig. 5a**), displacement was notably slow, resulting in a gradual increase in saturation and the longest breakthrough time. This behavior was attributed to the high resistance encountered by non-wetting water as it struggled to invade the pore network and overcome significant capillary entry pressures. The hydrophobic case exhibited a maximum water saturation of 44% and demanded the highest injection pressure of 62 mbar. The mixed-wettability scenarios demonstrated how hydrophilic pore throats influenced flow dynamics. In the Mix-Low case (**Fig. 5b**), where only 20% of the pore throats were hydrophilic, the injection pressure dropped slightly to 54 mbar, but the saturation decreased to 33%. Conversely, the Mix-High scenario (**Fig. 5c**), with hydrophilic throats well-distributed throughout the matrix, exhibited significantly improved dynamics. The pressure required for injection decreased to 47 mbar, and the water saturation increased to 45%, suggesting that the presence of sufficient hydrophilic pathways greatly enhanced pore accessibility and fluid distribution. In the hydrophilic case (**Fig. 5d**), water acted as the wetting phase, allowing capillary forces to dominate and facilitate fluid imbibition. This resulted in the fastest pore-filling process, with a rapid increase in saturation and achieving the highest breakthrough saturation of 55% across all scenarios. The associated pressure drop was the lowest at 33 mbar, demonstrating minimal resistance to flow due to the uniformly hydrophilic nature of the matrix.

The findings highlight the critical role of mixed wettability in determining fluid dynamics and sweep efficiency. Breakthrough times were observed to follow a consistent order: hydrophilic < Mix-High < Mix-Low < hydrophobic, correlating directly with the increasing proportion of hydrophilic regions in the porous model. The displacement process displayed periods of stagnation, evident in the saturation curves, characterized by static menisci and pressure build-ups. These phases were followed by abrupt increases in saturation, corresponding to distinct pore-filling events unique to each scenario. The saturation achieved in the Mix-High scenario fell between the hydrophilic and hydrophobic cases, whereas Mix-Low demonstrated lower saturation than both extremes. This challenges conventional assumptions, highlighting that mixed-wet systems exhibit unique and complex dynamics rather than merely acting as intermediates between uniform wettability extremes. These findings underscore the critical importance of both the spatial distribution and the overall proportion of hydrophilic pore throats in optimizing fluid displacement, particularly in mixed-wettability systems.

Conclusions

This study systematically investigated the influence of wettability on immiscible fluid displacement within porous media, employing high-resolution microfluidic models to mimic rock-like structures with controlled wettability distributions. Five distinct micromodels were developed, encompassing hydrophilic, hydrophobic, and three mixed-wettability configurations, all sharing identical pore structures. Wettability alterations were selectively applied to narrow pore throats based on pore size distribution, resulting in varied mixed-wettability proportions and spatial distributions. The results revealed that mixed-wettability systems exhibit unique and complex dynamics, distinct from hydrophilic and hydrophobic extremes, with displacement efficiency and saturation behavior strongly influenced by the spatial distribution and proportion of hydrophilic pore throats. Mixed-wet scenarios demonstrated intricate invasion patterns, where localized wettability variations redirected fluid flow and impacted sweep efficiency. Dynamic pressures were found to be inversely proportional to the fraction of hydrophilic zones. The interplay between capillary and viscous forces in mixed-wet systems deviated from anticipated intermediate behavior, highlighting the role of altered pore throats in shaping fluid propagation. These findings offer critical insights into multiphase flow dynamics, advancing the understanding of fluid behavior in heterogeneous porous systems and informing practical applications in energy and environmental fields.

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